

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

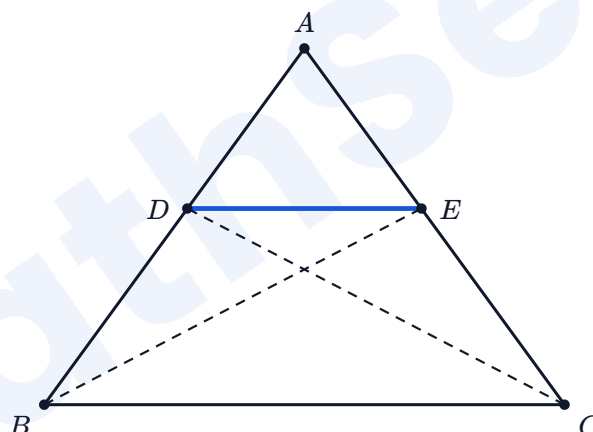
Chapter 6: Triangles

Previous year board questions, topic-wise: 13 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

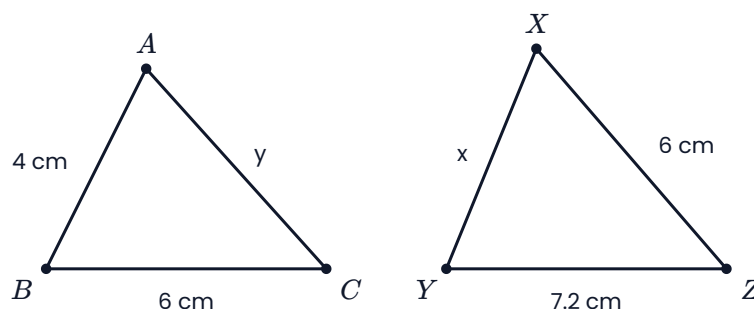
1 Basic Proportionality Theorem

1. In $\triangle ABC$, D lies on AB and E lies on AC with $DE \parallel BC$. If $AD = x$, $DB = x - 2$, $AE = x + 2$ and $EC = x - 1$, find x . **CBSE 2026, 2 marks**
2. State the Basic Proportionality Theorem and prove it. **CBSE 2026, 5 marks**
3. Prove that a line drawn parallel to one side of a triangle, cutting the other two sides at distinct points, divides those two sides in the same ratio. **CBSE 2024, 5 marks**

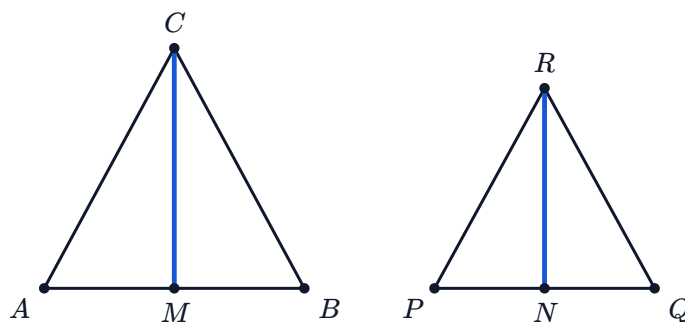


2 Criteria for similarity of triangles

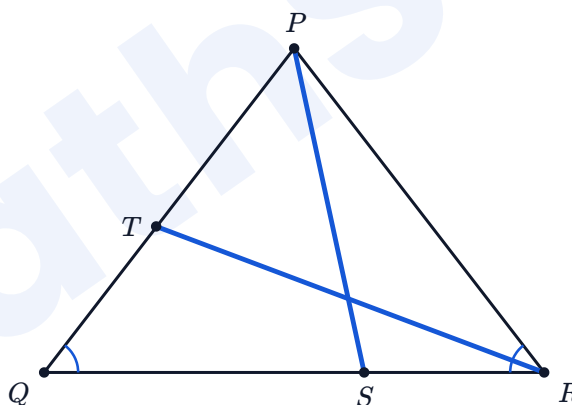
4. $\triangle ABC \sim \triangle DEF$ with $2AB = DE$ and $BC = 8$ cm. Find EF . (a) 4 cm (b) 8 cm (c) 12 cm (d) 16 cm **CBSE 2026, 1 mark**
5. $\triangle ABC \sim \triangle XYZ$ with $AB = 4$ cm, $BC = 6$ cm, $AC = y$ cm, $XY = x$ cm, $YZ = 7.2$ cm and $XZ = 6$ cm. Find x and y . **CBSE 2026, 2 marks**



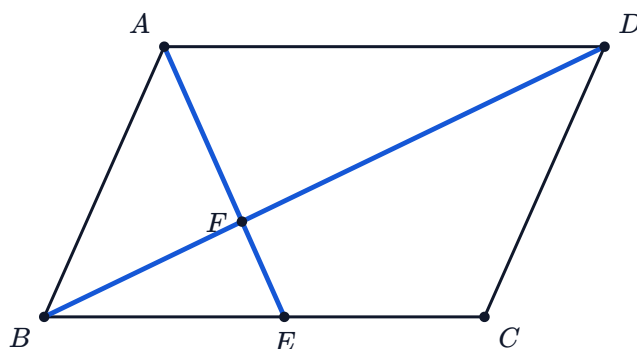
6. CM and RN are medians of $\triangle ABC$ and $\triangle PQR$, and $\triangle ABC \sim \triangle PQR$. Prove that (i) $\triangle AMC \sim \triangle PNR$ (ii) $\triangle CMB \sim \triangle RNQ$. **CBSE 2026, 5 marks**



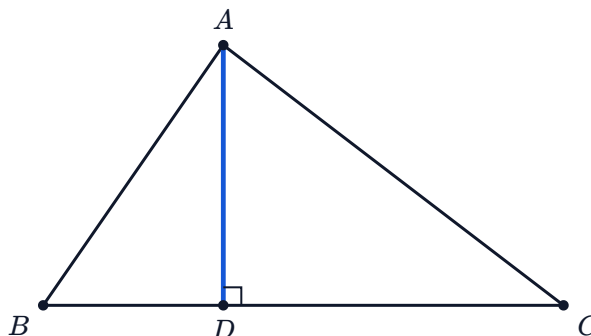
7. In $\triangle ABC$ and $\triangle DEF$, $\angle B = \angle E$, $\angle C = \angle F$ and $AB = 3DE$. Are the two triangles similar, congruent, both or neither? **CBSE 2025, 1 mark**
8. $\triangle ABC \sim \triangle PQR$ with $AB = 6$ cm, $BC = 4$ cm, $AC = 8$ cm and $PR = 6$ cm. Find $PQ + QR$. **CBSE 2025, 2 marks**
9. In $\triangle PQR$, S is a point on QR and T is a point on PQ . It is given that $\frac{QR}{QS} = \frac{QT}{PR}$ and $\angle PQR = \angle PRQ$. Prove that $\triangle PQS \sim \triangle TQR$. **CBSE 2025, 2 marks**



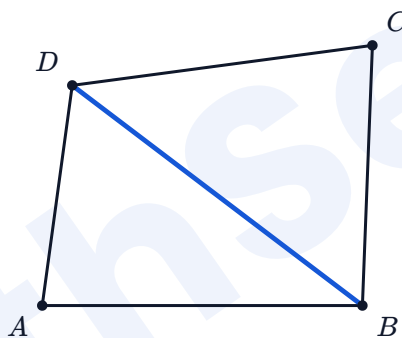
10. $ABCD$ is a parallelogram and E is a point on BC . The diagonal BD meets AE at F . Prove that $DF \times EF = FB \times FA$. **CBSE 2025, 5 marks**



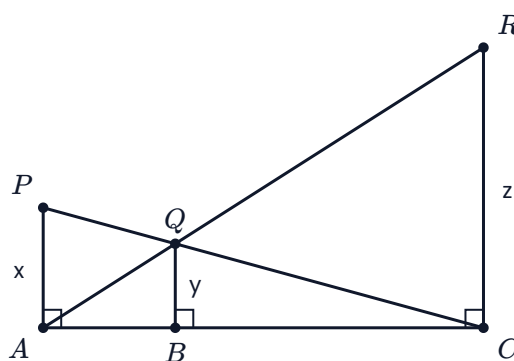
11. In $\triangle ABC$, D is a point on BC with $AD \perp BC$ and $AD^2 = BD \times DC$. Prove that $\angle BAC = 90^\circ$.
CBSE 2025, 5 marks



12. In quadrilateral $ABCD$, the diagonal BD bisects both $\angle B$ and $\angle D$. Prove that (i) $\triangle ABD \sim \triangle CBD$ (ii) $AB = BC$. CBSE 2024, 2 marks



13. PA , QB and RC are all perpendicular to the line AC , where Q is the point in which PC and AR meet. If $AP = x$, $BQ = y$ and $CR = z$, prove that $\frac{1}{x} + \frac{1}{z} = \frac{1}{y}$. CBSE 2024, 5 marks



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